# Graphical user interfaces (GUI)

- Tkinter

**primitive_calculator.py**

```python
accumulator = 0

while True:
    print('Accumulator:', accumulator)
    print('Select:')
    print('  1: clear')
    print('  2: add')
    print('  3: subtract')
    print('  4: multiply')
    print('  5: quit')

    choice = int(input('Choice: '))

    match choice:
        case 1: accumulator = 0
        case 2: accumulator += int(input('add: '))
        case 3: accumulator -= int(input('subtract: '))
        case 4: accumulator *= int(input('multiply by: '))
        case 5: break
```

**Python shell**

```
Accumulator: 0
Select:
  1: clear
  2: add
  3: subtract
  4: multiply
  5: quit
Choice: 2
add: 10
Accumulator: 10
Select:
  1: clear
  2: add
  3: subtract
  4: multiply
  5: quit
Choice: 2
add: 15
Accumulator: 25
Select:
...
```

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```
def memoize(f):
    # answers[args] = f(*args)
    answers = {}
    def wrapper(*args):
        return answers[args]
```

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# Graphical user interfaces (GUI)

- Tkinter

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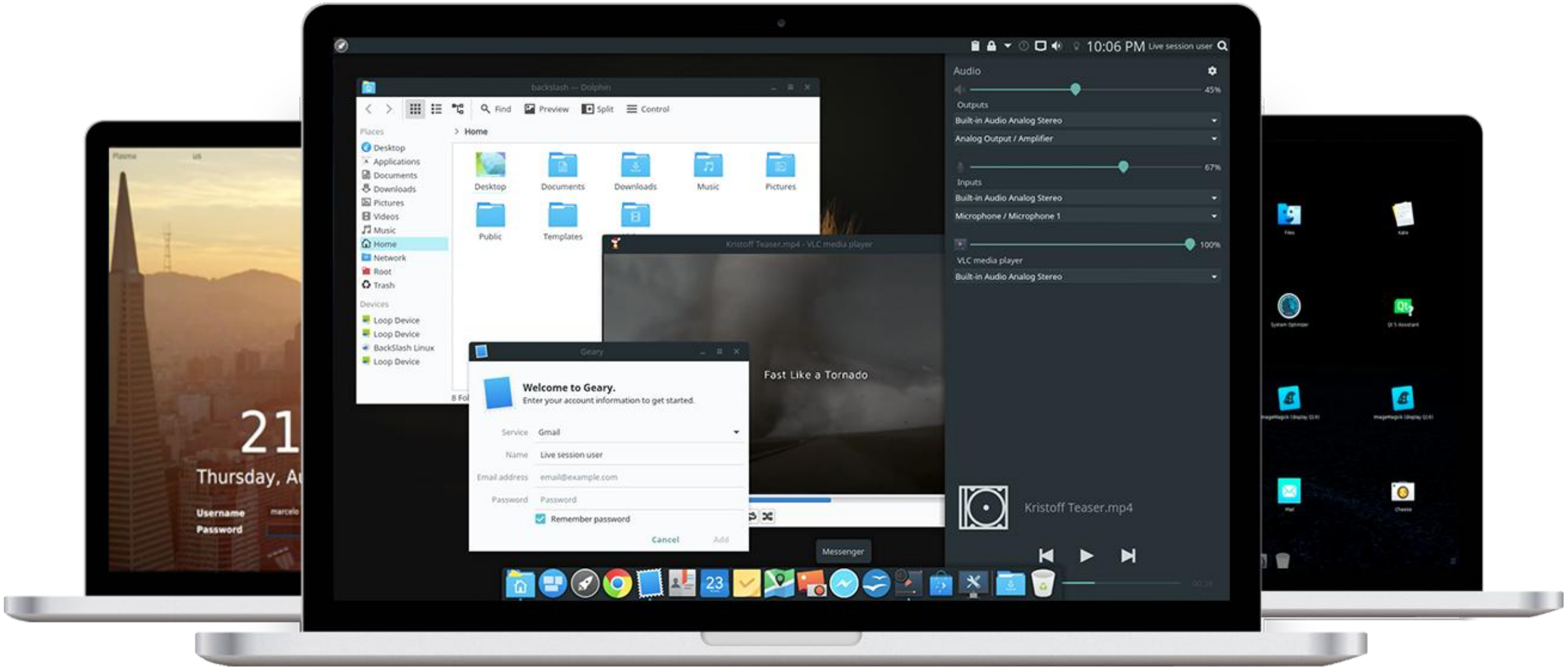



**BackSlash Linux GUI**
**www.backslashlinux.com**

# Python GUI's (Graphical Users Interfaces)

- There is a long list of GUI frameworks and toolkits, designer tools
  - we will only briefly look at Tkinter
- GUI are, opposed to a text terminal, easier to use, more intuitive and flexible
- Windows, icons, menus, buttons, scrollbars mouse / touch / keyboard interaction etc.
- Operating system (e.g. Windows, maxOS, iOS, Linux, Android) provides basic functionality in particular a window manager
- Writing GUI applications from scratch can be painful – frameworks try to provide all standard functionality

en.wikipedia.org/wiki/Colossal_Cave_Adventure

wiki.python.org/moin/GuiProgramming

# Tkinter

- “Tkinter is Python's de-facto standard GUI (Graphical User Interface) package. It is a thin object-oriented layer on top of Tcl/Tk.”
- “Tcl is a high-level, general-purpose, interpreted, dynamic programming language.”
- “Tk is a free and open-source, cross-platform widget toolkit that provides a library of basic elements of GUI widgets for building a graphical user interface (GUI) in many programming languages.”
- “The popular combination of Tcl with the Tk extension is referred to as Tcl/Tk, and enables building a graphical user interface (GUI) natively in Tcl. Tcl/Tk is included in the standard Python installation in the form of Tkinter.”

en.wikipedia.org

# Terminology

- widgets (e.g. buttons, editable text fields, labels, scrollbars, menus, radio buttons, check buttons, canvas for drawing, frames...)
- events (e.g. key press, mouse click, mouse entering/leaving, resizing windows, redraw requests, ...)
- listening (application waits for events to be fired)
- event handler (a function whose purpose is to handle an event, many triggered by user or OS/Window manager)
- geometry managers (how to organize widgets in a window: Tkinter *pack*, *grid*, *place*)

# docs.python.org/3/library/tk.html

*“tkinter is also famous for having an outdated look and feel”*

- Comes with Python
- Alternatives PySide, PyQt, Kivy, …

# Welcome example

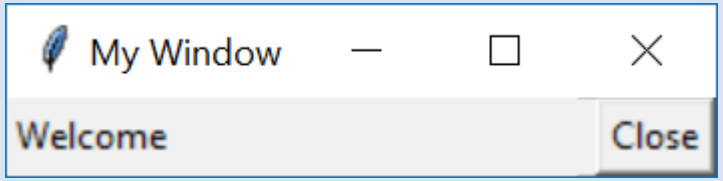
**welcome.py**

```
import tkinter
root = tkinter.Tk()  # root window
def do_quit():  # event handler for 'Close' button
    root.destroy()
root.title('Tkinter Welcome GUI')
label = tkinter.Label(root, text='Welcome to Tkinter', background='yellow',
                      anchor=tkinter.SE, font=('Helvetica', '24', 'bold italic'),
                      padx=10, pady=10)
label.pack(side=tkinter.LEFT, fill=tkinter.BOTH, expand=True)
                                          # parent window
close_button = tkinter.Button(root, text='Close', command=do_quit)
close_button.pack(side=tkinter.RIGHT)
tkinter.mainloop()  # loop until all windows are closed/destroyed
```

# Welcome example (class)

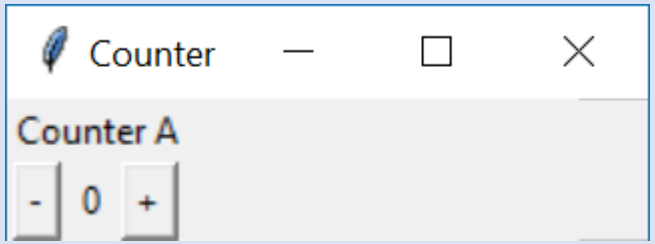
**welcome_class.py**

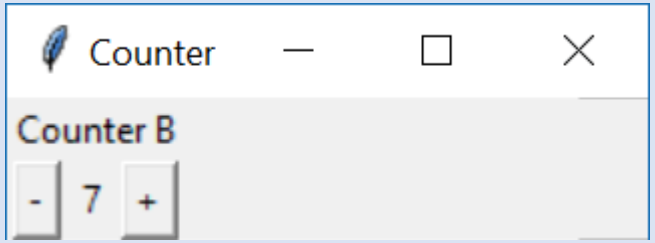
```
import tkinter

class Welcome:
    def do_quit(self):  # event handler for 'Close'
        self.root.destroy()

    def __init__(self, window_title):
        self.root = tkinter.Tk()
        self.root.title(window_title)

        self.label = tkinter.Label(self.root, text='Welcome')
        self.label.pack(side=tkinter.LEFT)

        self.close_button = tkinter.Button(self.root, text='Close', command=self.do_quit)
        self.close_button.pack(side=tkinter.RIGHT)

Welcome('My Window')

tkinter.mainloop()
```



## increment.py (part I)

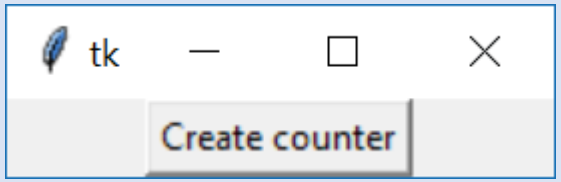
```
import tkinter

class Counter:
    def do_quit(self):
        self.root.destroy()

    def add(self, x):
        self.counter += x
        self.count.set(self.counter)

    def __init__(self, message):
        self.counter = 0

        self.root = tkinter.Toplevel()  # new window
        self.root.title('Counter')

        self.label = tkinter.Label(self.root, text=message)
        self.label.grid(row=0, columnspan=3)

        self.minus_button = tkinter.Button(self.root, text='-', command=lambda: self.add(-1))
        self.minus_button.grid(row=1, column=0)

        self.count = tkinter.IntVar()
        self.count_label = tkinter.Label(self.root, textvariable=self.count)
        self.count_label.grid(row=1, column=1)

        self.plus_button = tkinter.Button(self.root, text='+', command=lambda: self.add(+1))
        self.plus_button.grid(row=1, column=2)
```



**increment.py (part II)**

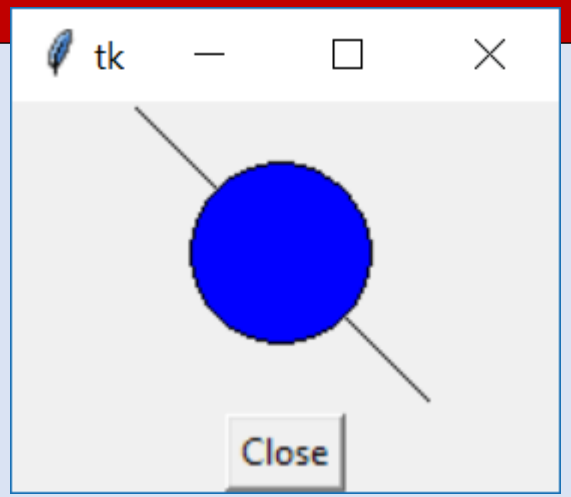
```
class Counter_app:
    def __init__(self):
        self.counters = 0

        self.root = tkinter.Tk()

        self.create = tkinter.Button(self.root, text='Create counter', command=self.new_counter)
        self.create.pack()

    def new_counter(self):
        Counter('Counter ' + chr(ord('A') + self.counters))
        self.counters += 1

Counter_app()

tkinter.mainloop()
```

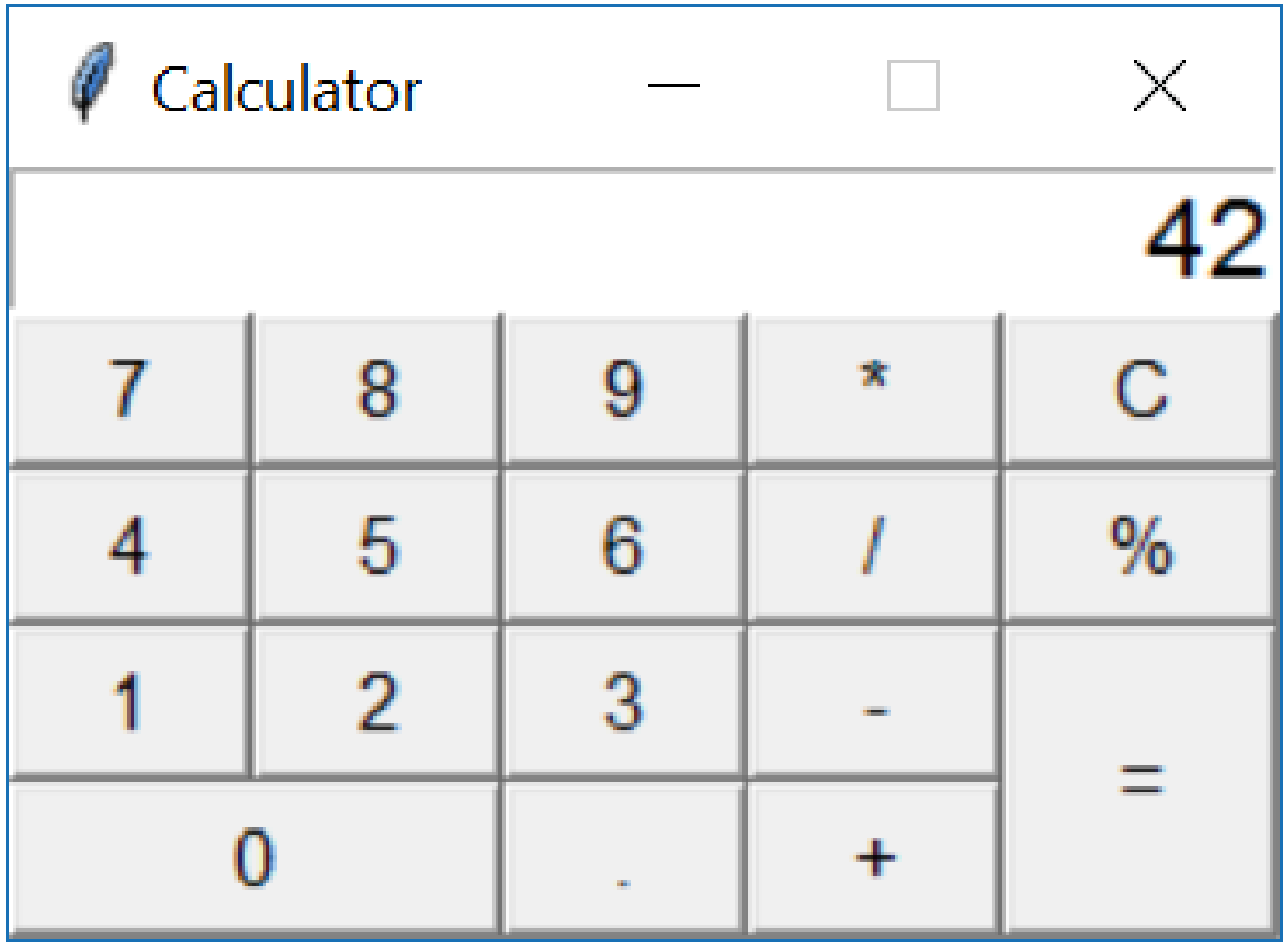


# Canvas

canvas.py

```
import tkinter

root = tkinter.Tk()

canvas = tkinter.Canvas(root, width=100, height=100)
canvas.pack()
canvas.create_line(0, 0, 100, 100)
canvas.create_oval(20, 20, 80, 80, fill='blue')

close = tkinter.Button(root, text='Close', command=root.destroy)
close.pack()

tkinter.mainloop()
```




Calculator
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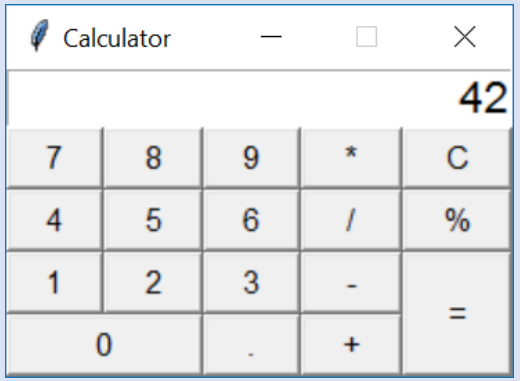
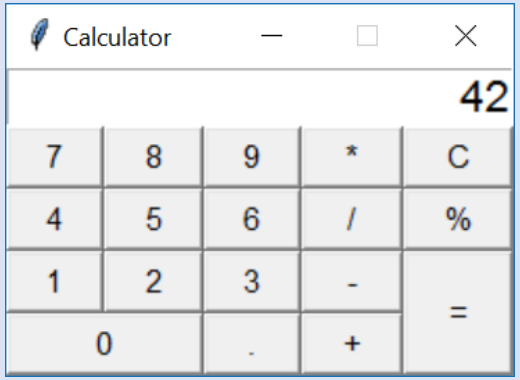
**calculator.py (Part I)**

```
import tkinter
from tkinter import messagebox

class Calculator:
    def __init__(self, root):
        self.root = root

        self.display = tkinter.Entry(self.root, font=('Helvetica', 16), justify=tkinter.RIGHT)
        self.display.insert(0, '0')
        self.display.grid(row=0, column=0, columnspan=5)  # grid = geometry manager

        self.button(1, 0, '7')
        self.button(1, 1, '8')
        self.button(1, 2, '9')
        self.button(1, 3, '*')
        self.button(1, 4, 'C', command=self.clearText)  # 'C' button
        self.button(2, 0, '4')
        self.button(2, 1, '5')
        self.button(2, 2, '6')
        self.button(2, 3, '/')
        self.button(2, 4, '%')
        self.button(3, 0, '1')
        self.button(3, 1, '2')
        self.button(3, 2, '3')
        self.button(3, 3, '-')
        self.button(3, 4, '=', rowspan=2, command=self.calculateExpression)  # '=' button
        self.button(4, 0, '0', columnspan=2)
        self.button(4, 2, '.')
        self.button(4, 3, '+')
```



## calculator.py (Part II)

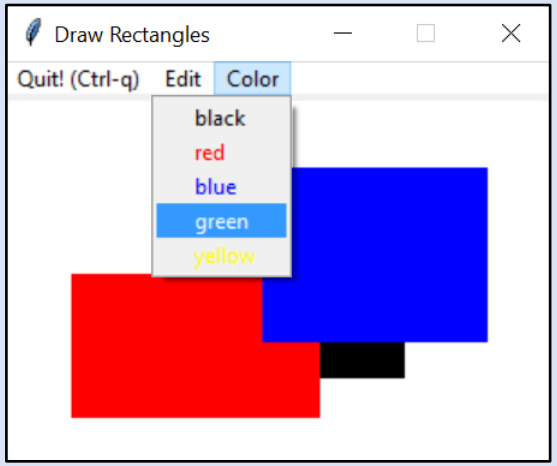
```
    def button(self, row, column, text, command=None, columnspan=1, rowspan=1):
        if command is None:
            command = lambda: self.appendToDisplay(text)
        B = tkinter.Button(self.root, font=('Helvetica', 11), text=text, command=command)
        B.grid(row=row, column=column, rowspan=rowspan, columnspan=columnspan, sticky='NWNESWSE')

    def clearText(self):
        self.replaceText('0')

    def replaceText(self, text):
        self.display.delete(0, tkinter.END)
        self.display.insert(0, text)

    def appendToDisplay(self, text):
        if self.display.get() == '0':
            self.replaceText(text)
        else:
            self.display.insert(tkinter.END, text)

    def calculateExpression(self):
        expression = self.display.get().replace('%', '/ 100')
        try:
            result = eval(expression)  # DON'T DO THIS !!!
            self.replaceText(result)
        except:
            messagebox.showwarning('Message', 'Invalid expression')

root = tkinter.Tk()
root.title('Calculator')
root.resizable(0, 0)  # disallow resizing and maximizing the window

Calculator(root)

tkinter.mainloop()
```



# Creating a menu

**rectangles.py**

```
class Rectangles:
    Colors = ['black', 'red', 'blue', 'green', 'yellow']

    def create_menu(self):
        menubar = tkinter.Menu(self.root)
        menubar.add_command(label='Quit! (Ctrl-q)', command=self.do_quit)

        editmenu = tkinter.Menu(menubar, tearoff=0)
        editmenu.add_command(label='Clear', command=self.clear_all)
        editmenu.add_command(label='Delete last (Ctrl-z)', command=self.delete_last_rectangle)

        colormenu = tkinter.Menu(menubar, tearoff=0)
        for color in self.Colors:  # list of color names
            colormenu.add_command(label=color,
                                  foreground=color,
                                  command=self.get_color_handler(color))

        menubar.add_cascade(label='Edit', menu=editmenu)
        menubar.add_cascade(label='Color', menu=colormenu)
        self.root.config(menu=menubar)  # Show menubar

    def get_color_handler(self, color):
        return lambda : self.set_color(color)

    def set_color(self, color):
        self.current_color = color
...
```

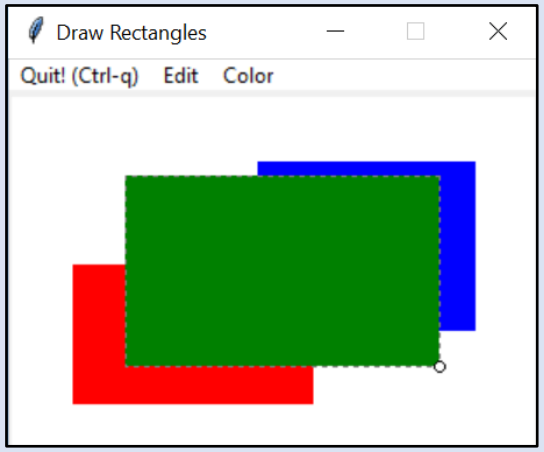


# Binding key and mouse events

- Whenever a key is pressed, mouse button is pressed/released, mouse is moved, mouse enters/leaves objects etc. events are triggered that can be bound to call a user defined event handler

**rectangles.py (continued)**

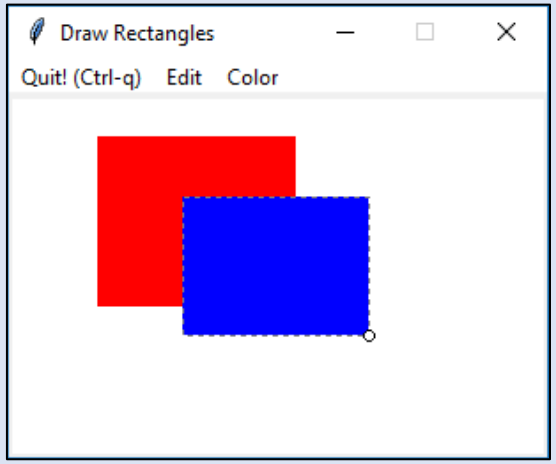
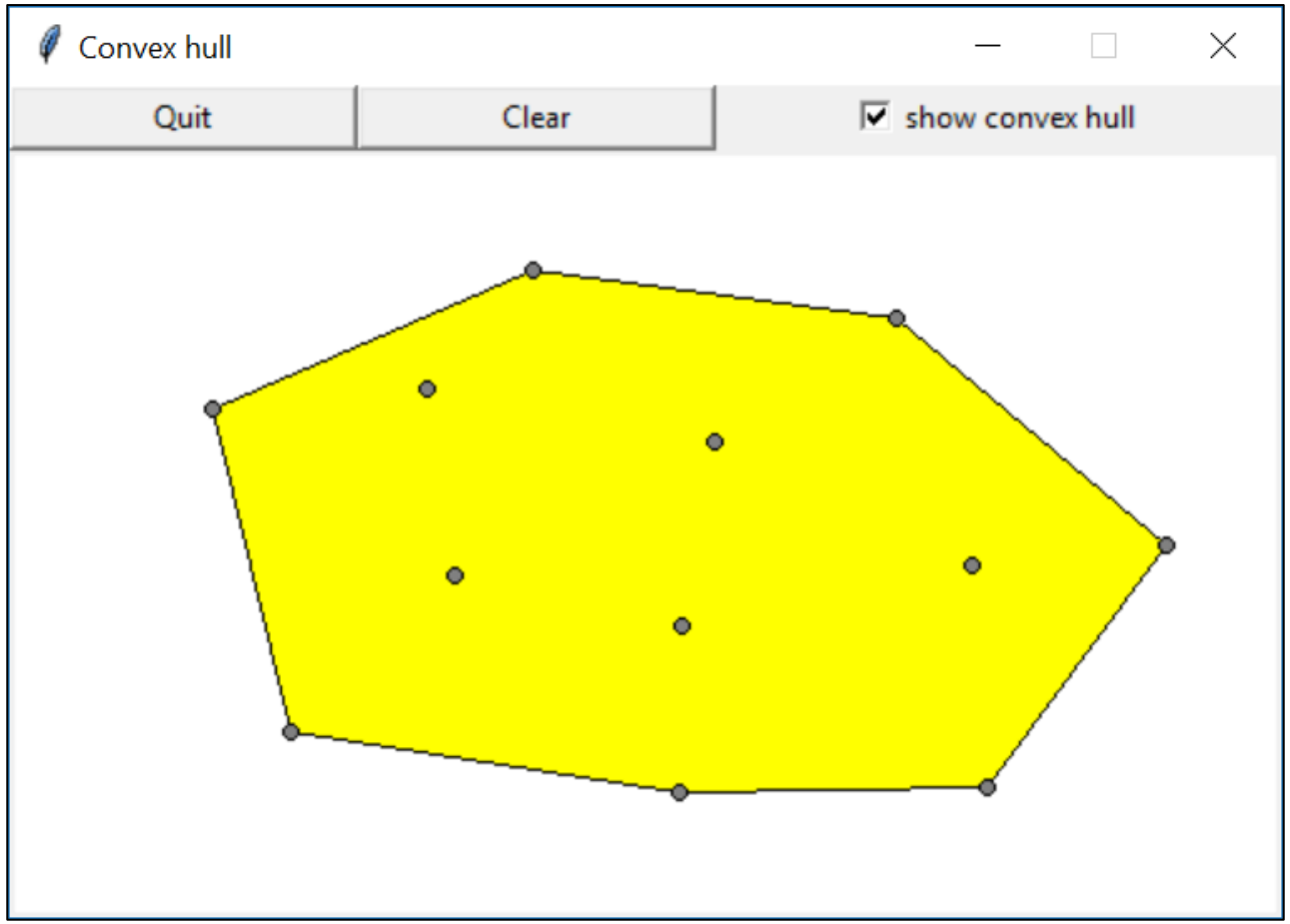
```
...
  self.root = tkinter.Tk()
  self.root.bind('<Control-q>', self.do_quit)
  self.root.bind('<Control-z>', self.delete_last_rectangle)
...
  self.canvas = tkinter.Canvas(self.root, width=300, height=200,
                               background='white')
  self.canvas.bind('<Button-1>', self.create_rectangle_start)
  self.canvas.bind('<B1-Motion>', self.create_rectangle_mouse_move)
  self.canvas.bind('<ButtonRelease-1>', self.create_rectangle_end)
...
```



# Handling mouse events

**rectangles.py (continued)**

```
def create_rectangle_start(self, event):
    radius = 3
    x, y = event.x, event.y
    self.top_pos = (x, y)
    self.bottom_pos = (x, y)
    self.rectangle = self.canvas.create_rectangle(x, y, x, y,  # top-left = bottom-right
                                                  fill=self.current_color, width=1, outline='grey', dash=(3, 5))
    self.corner = self.canvas.create_oval(x - radius, y - radius, x + radius, y + radius, fill='white')

def create_rectangle_mouse_move(self, event):
    if self.corner:
        x, y = event.x, event.y
        x_, y_ = self.bottom_pos
        self.bottom_pos = (x, y)
        self.canvas.coords(self.rectangle, *self.top_pos, *self.bottom_pos)
        self.canvas.move(self.corner, x - x_, y - y_)

def create_rectangle_end(self, event):
    if self.corner:
        self.canvas.delete(self.corner)
        self.corner = None
        if self.bottom_pos != self.top_pos:
            self.rectangles.append(self.rectangle)
            self.canvas.itemconfig(self.rectangle, width=0)
        else:  # empty rectangle, skip
            self.canvas.delete(self.rectangle)
        self.rectangle = None
```



# Exercise 25.1 (convex hull GUI)